

Complexity & Algorithms, Spring 2026

More lower
bounds



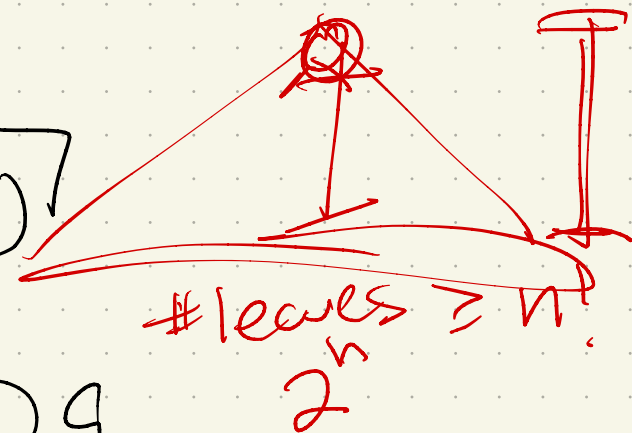
Next topics:

✓	Intro algorithms: Recursion, Dynamic Programming, and Greedy algorithms	11 respondents	61 %	✓
✓	Intro to graph algorithms: BFS/DFS, MST, and Shortest paths	11 respondents	61 %	
✓	Minimum spanning trees and shortest paths in graphs	9 respondents	50 %	
→	Flows and cuts on graphs, plus applications	3 respondents	17 %	
✓	Linear Programming (algorithms, duality, and applications)	6 respondents	33 %	
	Hashing/streaming/sketching data	7 respondents	39 %	
- ✗	Intro to randomized algorithms	9 respondents	50 %	
✓	Advanced data structures (Union-find, Fibonacci heaps, splay trees)	5 respondents	28 %	
✓	Approximation algorithms	10 respondents	56 %	
- ✗	Fast Fourier Transforms	7 respondents	39 %	
	Computational Geometry algorithms	5 respondents	28 %	
→ ✓	Lower bounds for problems - adversarial analysis	8 respondents	44 %	
- ✗	Dimensionality reduction - JL Lemma and SVDs	6 respondents	33 %	

Gaps in problem :

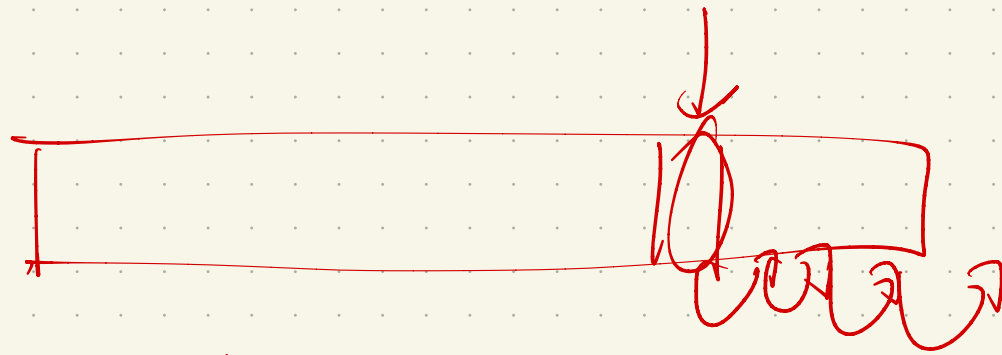
Sorting: $T(n) \geq \lceil \log_2(n!) \rceil$

for ~~$n=12$~~ , $T(n) \geq \underline{29}$



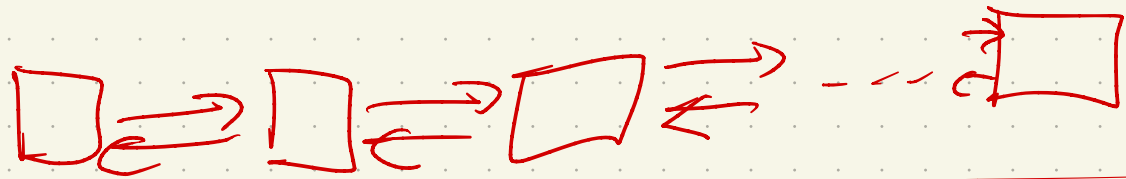
But : merge sort uses 30
comparisons

Can we get better lower bound?
no actual tree can have
depth 29



sorted

per insertion: $\lceil \log_2 n \rceil$ to find location
 $O(n)$ to move things



A new problem : Keister 1953

A function $f: \{0, 1, \dots, n-1\} \rightarrow \mathbb{R}$ is
unimodal if:

- $\exists$ a unique maximum x^*
- no two successive values are equal $f(i) \neq f(i+1)$

In other words:



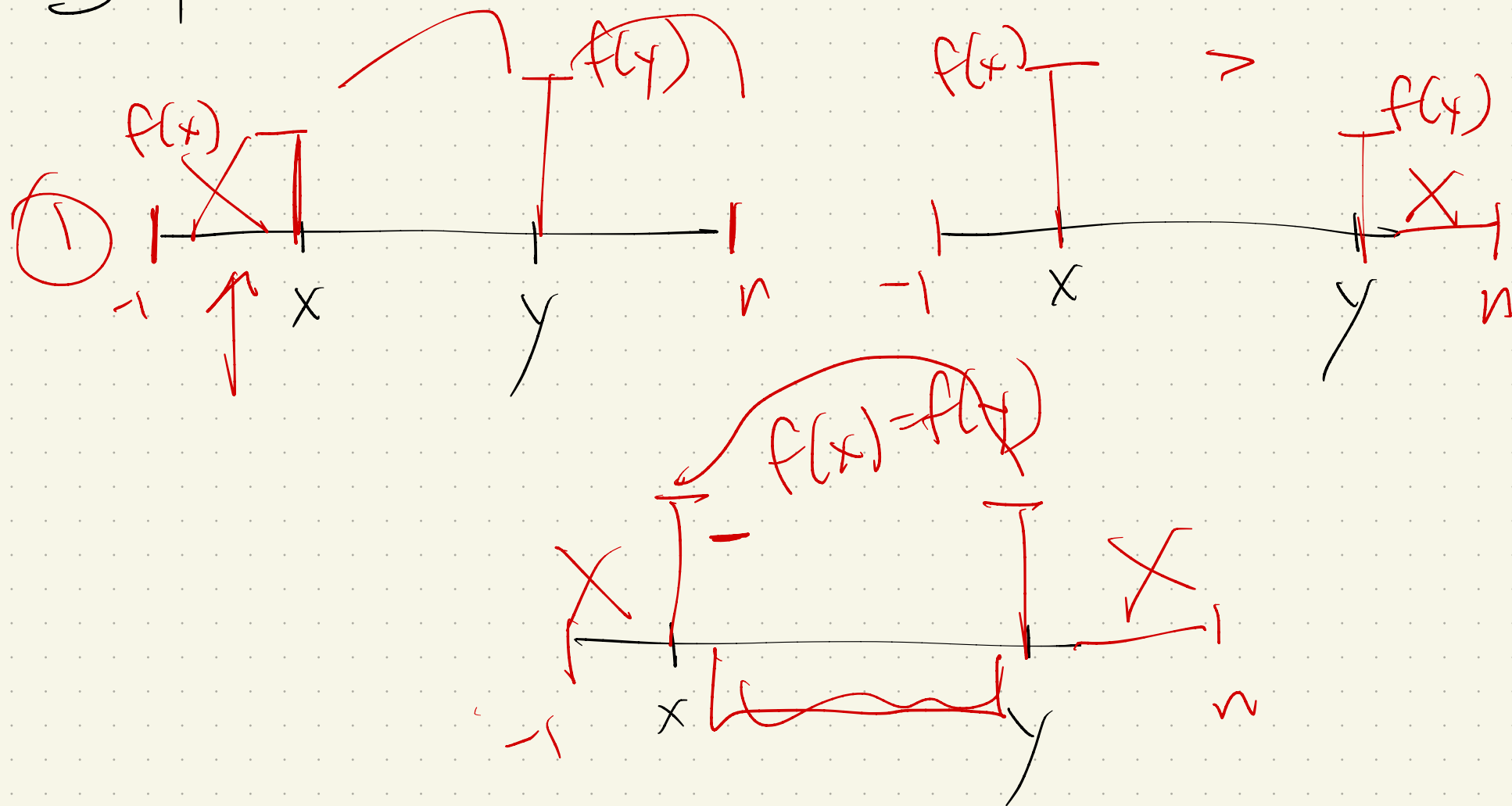
$f(0) < \dots < f(x^*-1) < f(x^*) > f(x^*+1) > \dots > f(n-1)$

1st *nth*

Goal: find x^* using as few function evaluations as possible

Suppose we know $f(x)$ & $f(y)$:

3 possibilities:



Turn this into decision tree:

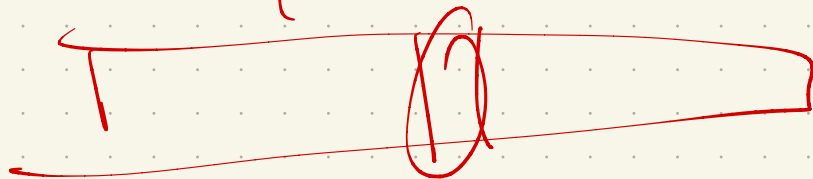
Pinning interval $[i, j]$:

an interval containing x^*

Initially: $[0, n-1]$, & our decision tree should shrink it.

Ideas for an algorithm?

very binary search!



Make this precise: $T(n) \leq 2^{\lceil \log_2 n \rceil}$

① Query powers of 2:

$$f(2) \leq f(4) \leq f(8) \leq \dots \leq f(2^{k-1}) < f(2^k)$$

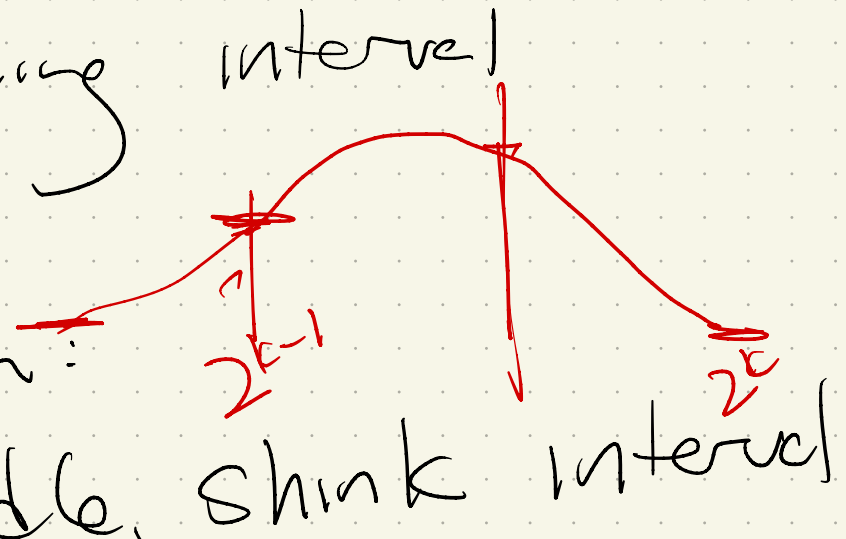
Stop when $f(2^k) < \underline{f(2^{k+1})}$

$\Rightarrow$ know x^* is in $[2^k, 2^{k+1}]$

How many queries? k queries

$$2^k \leq n \Rightarrow k \leq \lceil \log_2 n \rceil$$

② Now, have a pinning interval
 $[2^{k-1}, 2^k]$
 Do binary search:
 Query middle, shrink interval
 by $\frac{1}{2}$

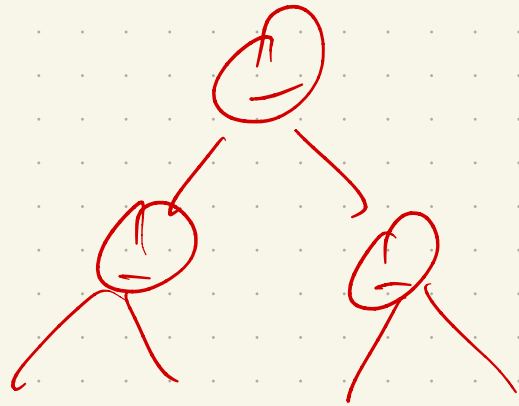


So # queries:

$$\lceil \log_2(2^k - 2^{k-1}) \rceil \leq \lceil \log_2 n \rceil$$

$$\text{total: } \leq 2 \lceil \log_2 n \rceil$$

Lower bound: $T(n) \geq c \cdot \log_2 n$?
How many leaves in decision tree?



n outputs

leaves

$$h \geq \lceil \log_2 n \rceil$$

How to get exact answer?

Aside: "Little Birdie Principle"

Extra information cannot
increase complexity
of problem.

Here: decreasing pinning
interval can only help me
→ function $T(n)$ is non-decreasing

To get a tighter bound:
 Invert the problem

Let $N(t)$ = size of largest
 pinning range where we can
 find x in $\leq t$ probes

Small t :

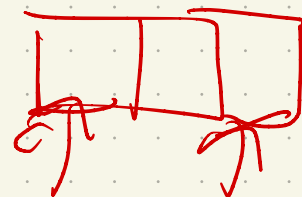
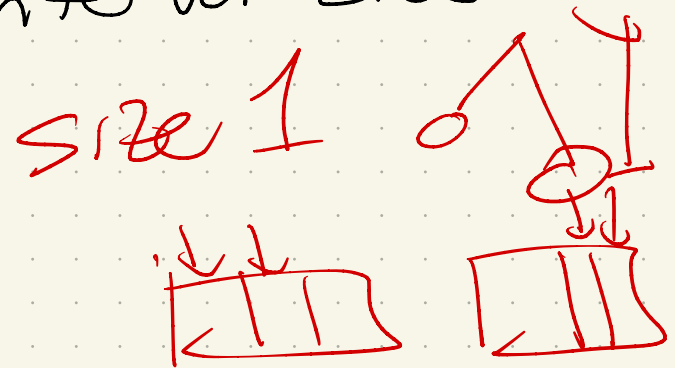
$t=0$ $N(0)=1$

$t=1$ $N(1)=1$

$t=2$ $N(2)=2$

$t=3$ $N(3)=4$

Interval size:



Our previous bounds:

$$t \leq 2 \lceil \log_2 n \rceil \quad T(n) \leq 2 \lceil \log_2 n \rceil$$

$$t/2 \leq \lceil \log_2 n \rceil$$

$$\Rightarrow N(t) \geq 2^{t/2}$$

$$2^{t/2} \leq n$$

$$T(n) \geq \lceil \log_2 n \rceil$$

$$\Rightarrow N(t) \leq 2^t$$

$$2^{t/2} \leq N(t) \leq 2^t \quad \text{gap!}$$

Theorem: $N(t) = F_{t+1} - 1$

where $F_k = k^{\text{th}}$ Fibonacci number.

Proof: Induction on t :

Base cases:

$$N(1) = 1$$

$$N(2) = 2$$

$$F_2 - 1 = 2 - 1 = 1$$

$$F_3 - 1 = 3 - 1 = 2$$

I.H: Suppose $N(k) = F_{k+1} - 1$
for $k < t$

With 2 queries,
 went from range of size n
 to range of size $N(t-1)$



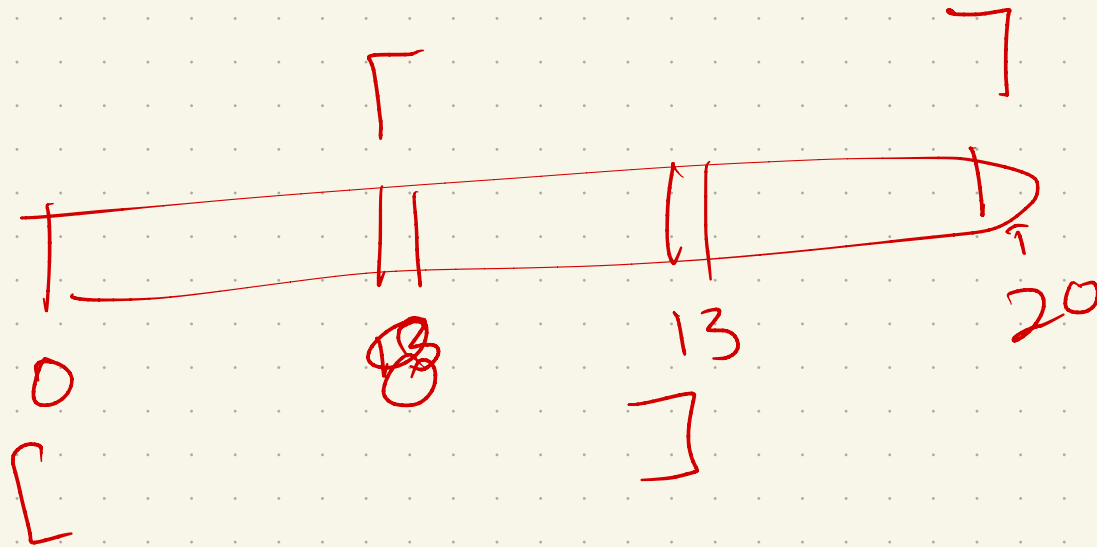
By IH: can do $N(t-1)$ using $t-1$ queries

$$\text{Total: } 2 + \cancel{(t-1)} = t+1 \quad \text{"}$$

$$(t-2) = t \quad \text{"}$$

b/c already
 did one of them

1, 1, 2, 3, 5, 8, 13, 21



Lower bound:

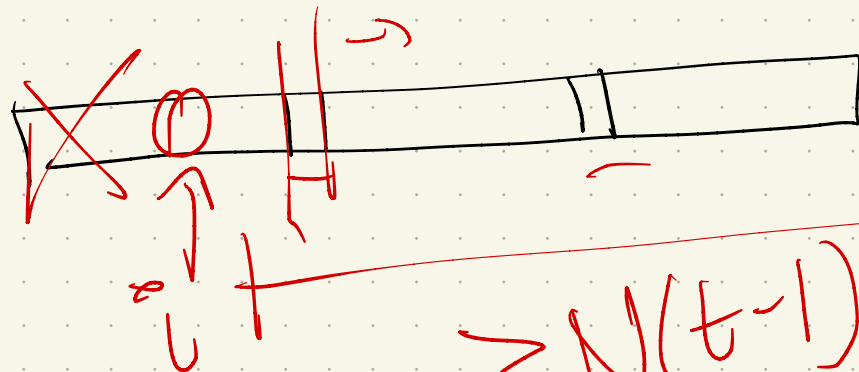
Suppose you probe at i & j

if $i = N(t-2)$ & $j = N(t-1)$

see prev slide

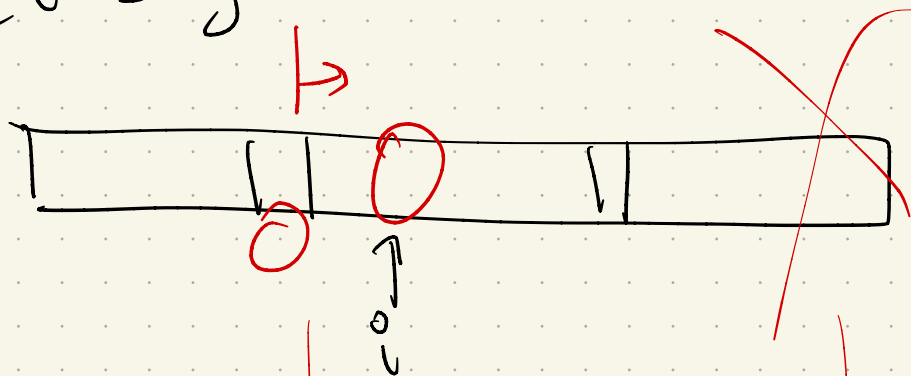
if $i < N(t-2)$:

adversary says " $f(i) < f(j)$ "



> $N(t-1)$
more than $t-1$
probes

if $i > N(t-2)$:



Say " $f(i) > f(j)$ "

and also shares " $x^* < i$ "

↳ can only help!

Then $i \geq N(t-2)$
need t or more